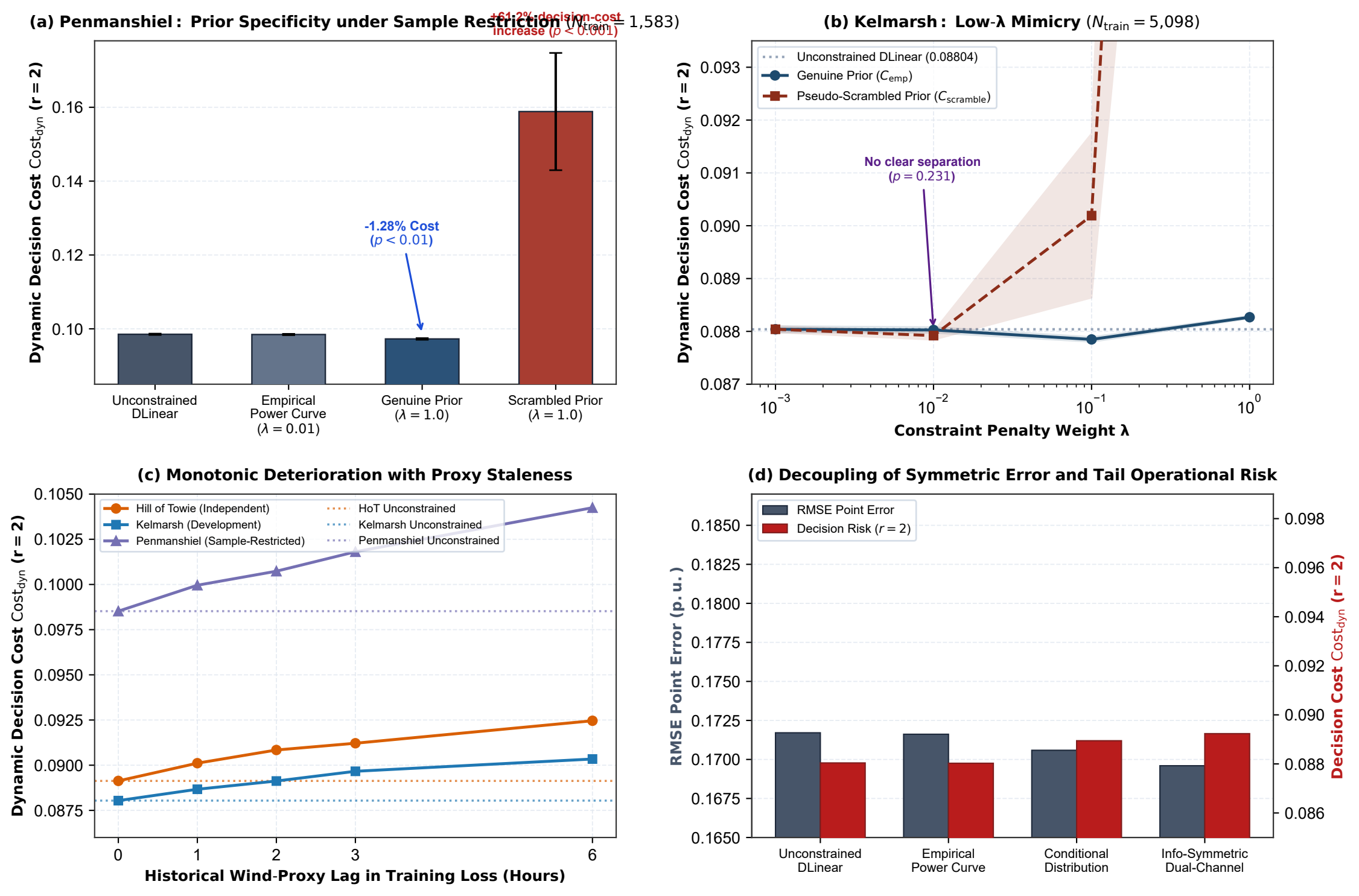

(a) Penmanshiel : Prior Specificity under Sample Restriction (N_{train} = 1,583)
+61.2% decision-cost increase ($p < 0.001$)
Dynamic Decision Cost $Cost_{dyn}$ (r = 2)
0.16
0.14
0.12
0.10
-1.28% Cost
($p < 0.01$)
Unconstrained DLinear
Empirical Power Curve ($\lambda = 0.01$)
Genuine Prior ($\lambda = 1.0$)
Scrambled Prior ($\lambda = 1.0$)
(b) Kelmarsh : Low-λ Mimicry (N_{train} = 5,098)
Unconstrained DLinear (0.08804)
Genuine Prior (C_{emp})
Pseudo-Scrambled Prior ($C_{scramble}$)
No clear separation
($p = 0.231$)
Dynamic Decision Cost $Cost_{dyn}$ (r = 2)
0.093
0.092
0.091
0.090
0.089
0.088
0.087
10^{-3}
10^{-2}
10^{-1}
10^{0}
Constraint Penalty Weight λ
(c) Monotonic Deterioration with Proxy Staleness
Hill of Towie (Independent)
Kelmarsh (Development)
Penmanshiel (Sample-Restricted)
HoT Unconstrained
Kelmarsh Unconstrained
Penmanshiel Unconstrained
Dynamic Decision Cost $Cost_{dyn}$ (r = 2)
0.1050
0.1025
0.1000
0.0975
0.0950
0.0925
0.0900
0.0875
0
1
2
3
6
Historical Wind-Proxy Lag in Training Loss (Hours)
(d) Decoupling of Symmetric Error and Tail Operational Risk
RMSE Point Error
Decision Risk (r = 2)
RMSE Point Error (p. u.)
0.1850
0.1825
0.1800
0.1775
0.1750
0.1725
0.1700
0.1675
0.1650
Decision Cost $Cost_{dyn}$ (r = 2)
0.098
0.096
0.094
0.092
0.090
0.088
0.086
Unconstrained DLinear
Empirical Power Curve
Conditional Distribution
Info-Symmetric Dual-Channel